Describing and Understanding Open Play: Post-cleaning

This follows on from `cleaning.qmd`, where I focused on understanding the participants and the general demographics of the participants. Here, I begin to understand the data, gaining more deeper insights and exploring the telemetry data of the qualified participants.

Prof. Paul Cairns is really interested in any potential differences between people with ADHD and other players. There are also well-being measures including a measure for depression - it could be interesting to see to what extent how people play interacts with how well they are doing more generally.

Demographics

First, I will get the general demographic descriptive statistics for the participants.

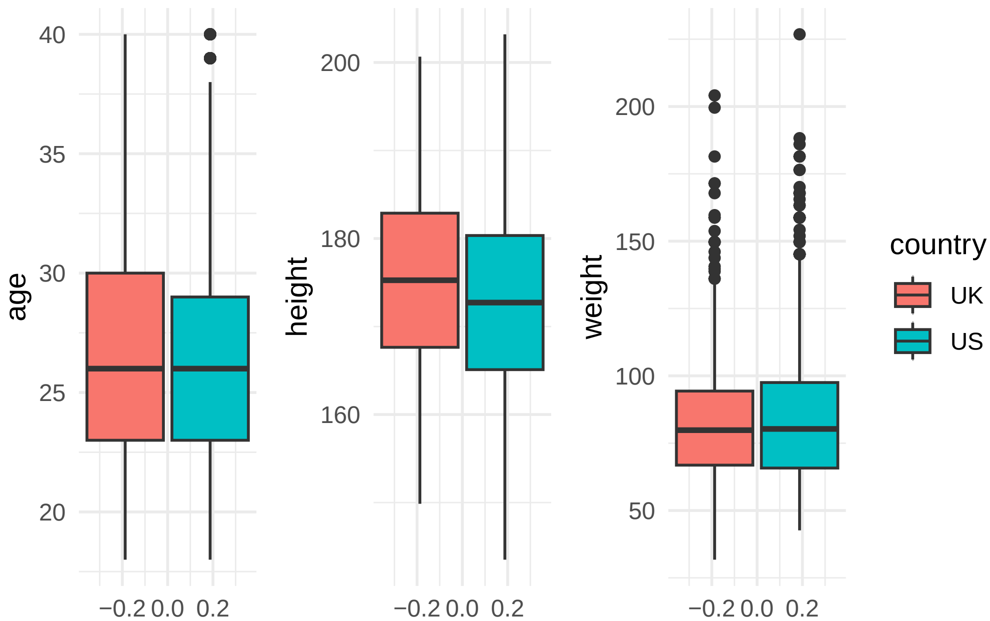
```
# create ADHD flag
intake_participants <- intake_participants |>
  mutate(
    adhd = case_when(
      !is.na(neuro_diag_adhd) ~ "Diagnosed ADHD",
      is.na(neuro_diag_adhd) & !is.na(neuro_iden_adhd) ~ "Self-Identified ADHD",
      TRUE ~ "Neurotypical"
    )
  )

demographic_vars <- c(
  "pid",
  "country",
  "geo_area",
  "gender",
  "height",
```

```r
  "weight",
  "age",
  "ethnicity",
  "edu_level",
  "employment",
  "political_party",
  "marital_status",
  "adhd"
)

intake_participants |>
  select(all_of(setdiff(demographic_vars, c("pid", "geo_area")))) |>
  mutate(across(where(is.character), as.factor)) |>
  skim() |> print()
```

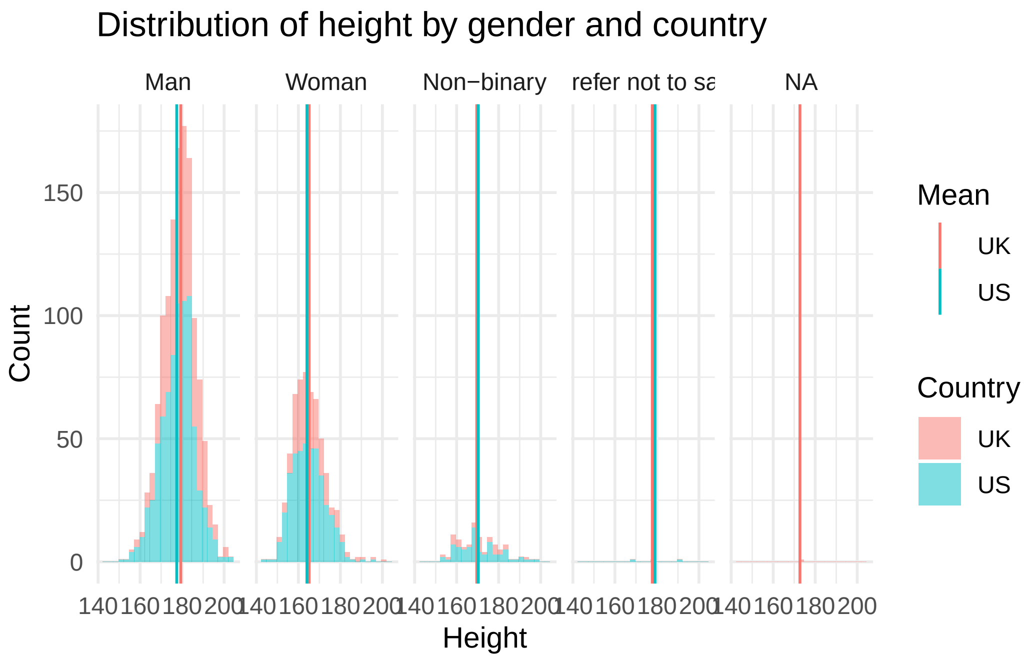
```
-- Data Summary ------------------------
                           Values
Name                       mutate(...)
Number of rows             1978
Number of columns          11

_______________________
Column type frequency:
  factor                   8
  numeric                  3

________________________
Group variables            None

-- Variable type: factor -------------------------------------------------------
  skim_variable   n_missing complete_rate ordered n_unique
1 country                 0         1     FALSE          2
2 gender                  1         0.999 FALSE          4
3 ethnicity               0         1     FALSE          9
4 edu_level               0         1     FALSE          8
5 employment              0         1     FALSE          8
6 political_party         0         1     FALSE         15
7 marital_status          0         1     FALSE          6
8 adhd                    0         1     FALSE          3
  top_counts
1 US: 1259, UK: 719
2 Man: 1282, Wom: 587, Non: 105, Pre: 3
3 Whi: 1366, Mul: 199, Asi: 159, Bla: 145
4 Bac: 607, Hig: 525, Hig: 432, Pos: 197
```

```
5 Wor: 865, Not: 374, Stu: 325, Wor: 284
6 Dem: 591, Ind: 428, Lab: 220, Rep: 178
7 Sin: 1419, Mar: 266, In : 243, Div: 25
8 Neu: 1475, Dia: 331, Sel: 172

-- Variable type: numeric ------------------------------------------------------
  skim_variable n_missing complete_rate mean    sd    p0   p25   p50   p75 p100
1 height                0             1 174.  10.3  144.  165.  175.  180.  203.
2 weight                0             1  83.6 24.5   31.8  65.8  79.8  95.2 227.
3 age                   0             1  26.7  4.97  18    23    26    30    40
  hist
1
2
3
```

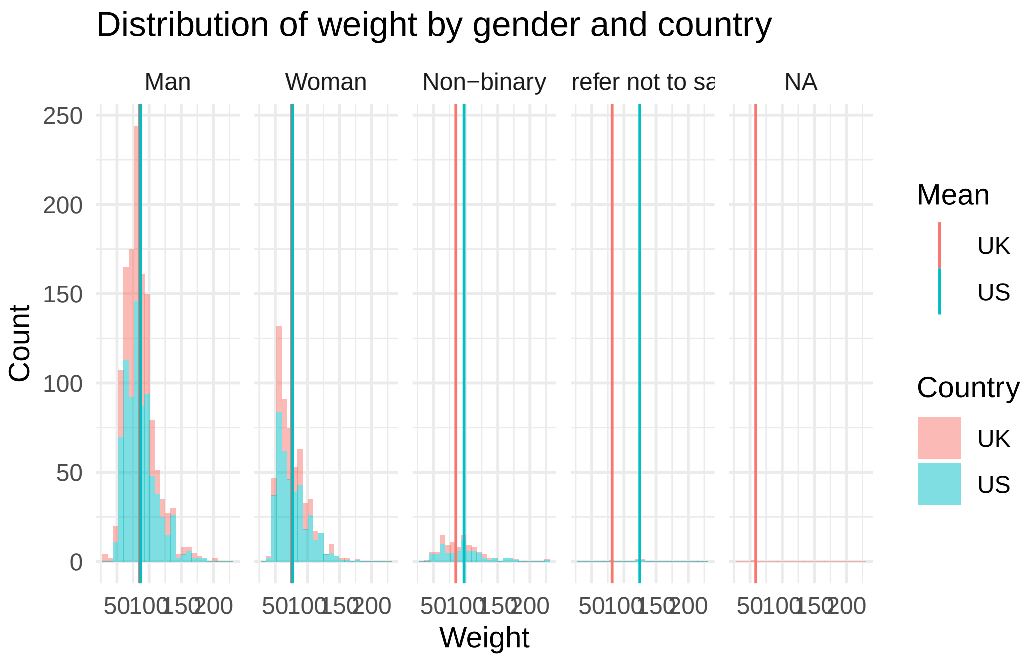
```r
# height and weight by age, gender, country
# height_weight_data <- intake_participants |>
#   select(height, weight, age, gender, country) |>
#   mutate(size)
mean_values <- intake_participants |>
  group_by(country, gender) |>
  summarise(
    height_mean = mean(height),
    weight_mean = mean(weight),
    age_mean = mean(age),
    .groups = "drop_last"
  )

age_box <- ggplot(intake_participants, aes(y = age, fill = country)) +
  geom_boxplot()
height_box <- ggplot(intake_participants, aes(y = height, fill = country)) +
  geom_boxplot()
weight_box <- ggplot(intake_participants, aes(y = weight, fill = country)) +
  geom_boxplot()

age_box + height_box + weight_box + plot_layout(guides = "collect")
```



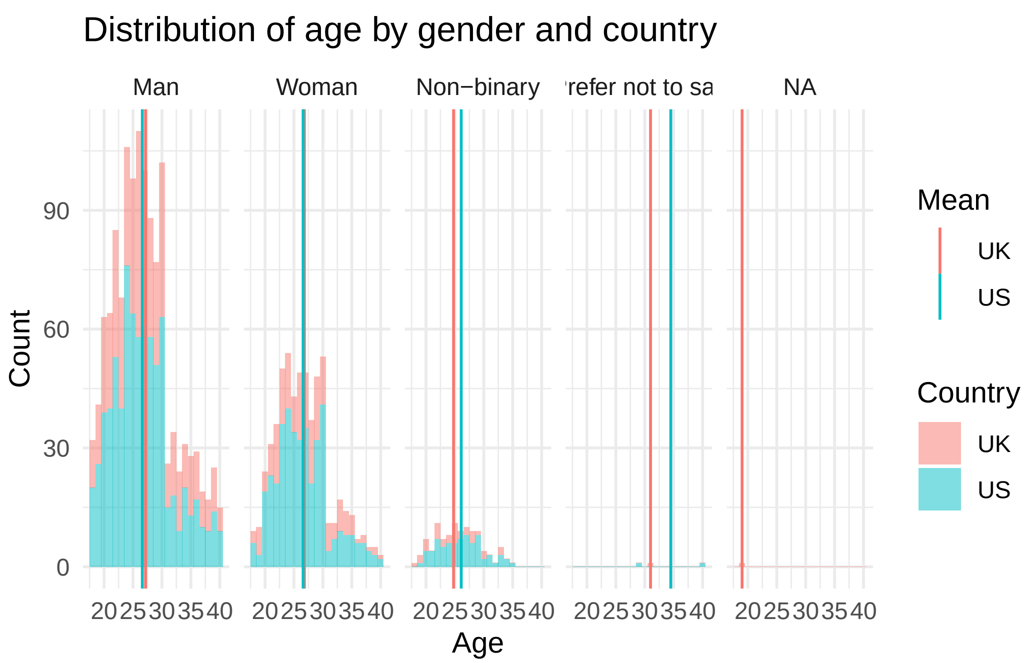

```
ggplot(intake_participants, aes(x = height, fill = country)) +
  geom_histogram(bins = 25, alpha = 0.5) +
  facet_grid(
    ~factor(gender, levels = c("Man", "Woman", "Non-binary", "Prefer not to say", "NA")),
    scales = "free_y"
  ) +
  geom_vline(data = mean_values, aes(xintercept = height_mean, colour = country)) +
  labs(
    x = "Height",
    y = "Count",
    fill = "Country",
    colour = "Mean",
    title = "Distribution of height by gender and country"
  )
```




```
ggplot(intake_participants, aes(x = weight, fill = country)) +
  geom_histogram(bins = 25, alpha = 0.5) +
  facet_grid(
    ~factor(gender, levels = c("Man", "Woman", "Non-binary", "Prefer not to say", "NA")),
    scales = "free_y"
  ) +
  geom_vline(data = mean_values, aes(xintercept = weight_mean, colour = country)) +
  labs(
    x = "Weight",
    y = "Count",
    fill = "Country",
    colour = "Mean",
    title = "Distribution of weight by gender and country"
  )
```




```
ggplot(intake_participants, aes(x = age, fill = country)) +
  geom_histogram(bins = 23, alpha = 0.5) +
  facet_grid(
    ~factor(gender, levels = c("Man", "Woman", "Non-binary", "Prefer not to say", "NA")),
    scales = "free_y"
  ) +
  geom_vline(data = mean_values, aes(xintercept = age_mean, colour = country)) +
  labs(
    x = "Age",
    y = "Count",
    fill = "Country",
    colour = "Mean",
    title = "Distribution of age by gender and country"
  )
```




```
# cross-tabs for adhd
addmargins(xtabs(~gender + adhd, data = intake_participants))
```

```
                   adhd
gender              Diagnosed ADHD Neurotypical Self-Identified ADHD  Sum
  Man                          170         1018                   94 1282
  Non-binary                    48           45                   12  105
  Prefer not to say              1            2                    0    3
  Woman                        111          410                   66  587
  Sum                          330         1475                  172 1977
```

```
addmargins(xtabs(~country + adhd, data = intake_participants))
```

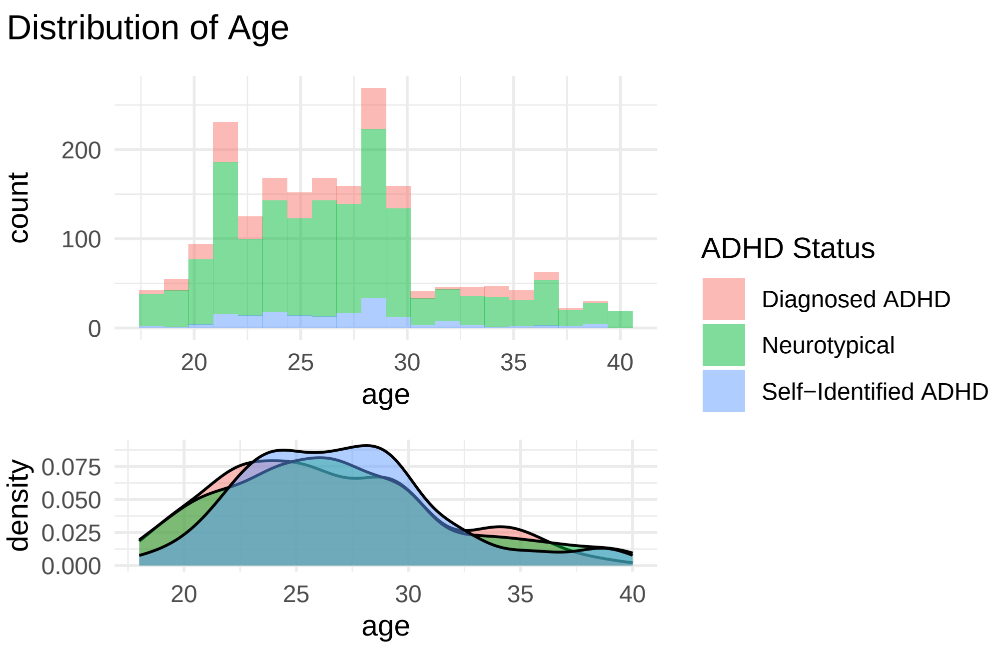
```
       adhd
country Diagnosed ADHD Neurotypical Self-Identified ADHD  Sum
    UK              73          570                   76  719
    US             258          905                   96 1259
    Sum            331         1475                  172 1978
```

```
addmargins(xtabs(~ethnicity + adhd, data = intake_participants))
```

```
                                           adhd
ethnicity                                   Diagnosed ADHD Neurotypical
  American Indian and Alaska Native                      1            7
  Asian                                                 14          133
  Black                                                 23          116
  Multiple                                              37          144
  Native Hawaiian and Other Pacific Islander             0            2
  Not reported                                          10           31
  Other                                                  7           42
  Prefer not to say                                      0            1
  White                                                239          999
  Sum                                                  331         1475
                                           adhd
ethnicity                                   Self-Identified ADHD  Sum
  American Indian and Alaska Native                            0    8
  Asian                                                       12  159
  Black                                                        6  145
  Multiple                                                    18  199
  Native Hawaiian and Other Pacific Islander                   0    2
  Not reported                                                 3   44
  Other                                                        4   53
  Prefer not to say                                            1    2
  White                                                      128 1366
  Sum                                                        172 1978
```

```r
# plot distributions with adhd
plot_data <- intake_participants |>
  mutate(adhd = as.factor(adhd))

adhd_age_density <- ggplot(plot_data, aes(x = age, fill = adhd)) +
  geom_density(alpha = 0.5, show.legend = TRUE) +
  labs(fill = "ADHD Status") +
  guides(fill = "none")

adhd_age_hist <- ggplot(plot_data, aes(x = age, fill = adhd)) +
  geom_histogram(bins = 20, alpha = 0.5, show.legend = TRUE) +
  labs(fill = "ADHD Status")

(adhd_age_hist / adhd_age_density) +
  plot_annotation(title = 'Distribution of Age') +
  plot_layout(
    guides = "collect",
```

```
    heights = c(2, 1)
  )
```

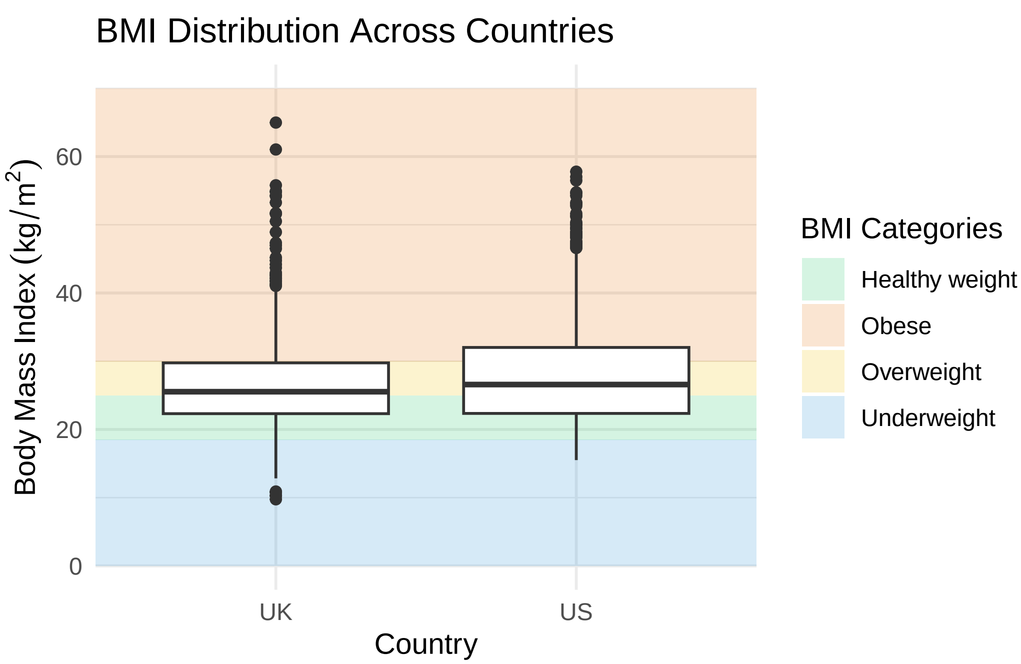


Similar (almost normal?) distributions of height. Male participants have slightly larger height than Female, but Non-binary appears to be spread almost more evenly. The country does not affect the distributions.

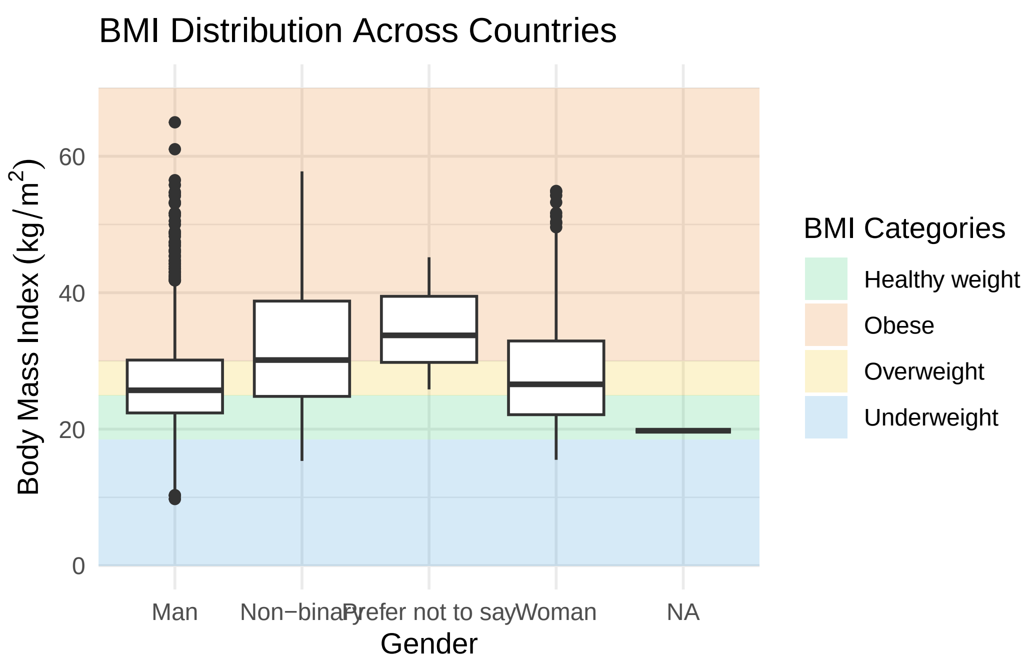
I would like to quicky check BMI:

```
# BMI calculator (we do have 18 year olds however)
intake_participants <- intake_participants |>
  mutate(
    bmi = (weight / ((height/100) ^ 2))
  )

bmi_categories <- data.table(
  category = c("Underweight", "Healthy weight", "Overweight", "Obese"),
  bmi_min = c(0, 18.5, 25.0, 30.0),
  bmi_max = c(18.5, 25.0, 30.0, max(intake_participants$bmi)+5), # set hard limit
  colour = c("#3498db", "#2ecc71", "#f1c40f", "#e67e22")
)
category_palette <- setNames(bmi_categories$colour, bmi_categories$category)

# BMI across countries
```

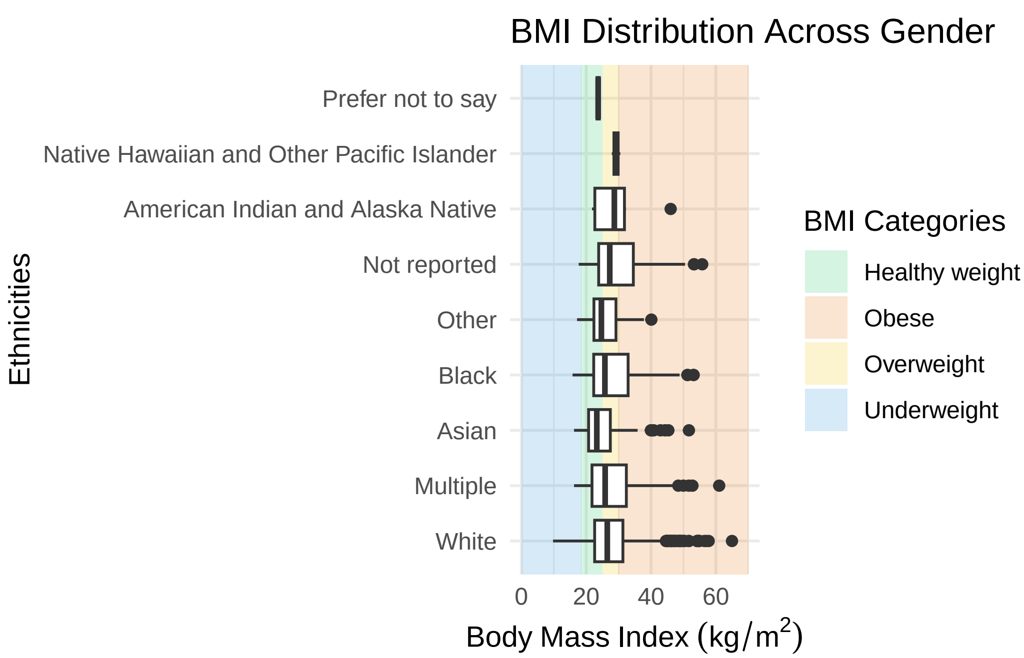
```r
ggplot() +
  geom_rect(
    data = bmi_categories,
    aes(xmin = -Inf, xmax = Inf, ymin = bmi_min, ymax = bmi_max, fill = category),
    alpha = 0.2) +
  geom_boxplot(
    data = intake_participants,
    aes(
      x = as.factor(country),
      y = bmi,
    ),
    fill = "white"
    ) +
  scale_fill_manual(values = category_palette, name = "BMI Categories") +
  labs(
    title = "BMI Distribution Across Countries",
    x = "Country",
    y = expression(Body ~ Mass ~ Index ~ (kg/m^2))
  ) +
  guides(fill = "legend", colour = "none")
```



```
# BMI across gender
ggplot() +
  geom_rect(
    data = bmi_categories,
    aes(xmin = -Inf, xmax = Inf, ymin = bmi_min, ymax = bmi_max, fill = category),
    alpha = 0.2) +
  geom_boxplot(
    data = intake_participants,
    aes(
      x = as.factor(gender),
      y = bmi,
    ),
    fill = "white"
    ) +
  scale_fill_manual(values = category_palette, name = "BMI Categories") +
  labs(
    title = "BMI Distribution Across Countries",
    x = "Gender",
    y = expression(Body ~ Mass ~ Index ~ (kg/m^2))
  ) +
  guides(fill = "legend", colour = "none")
```

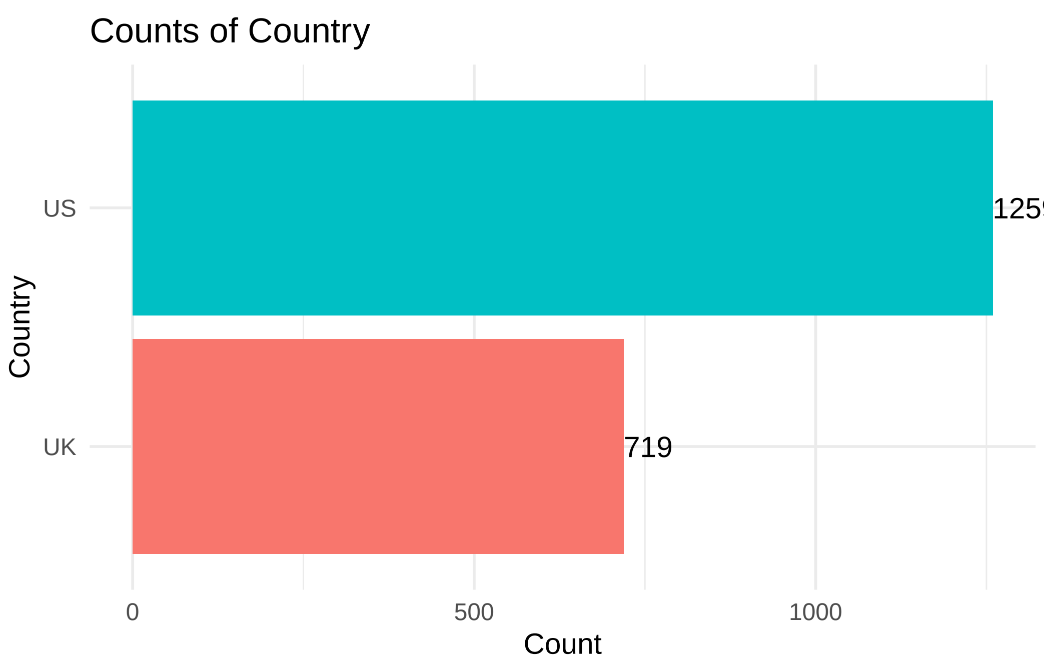


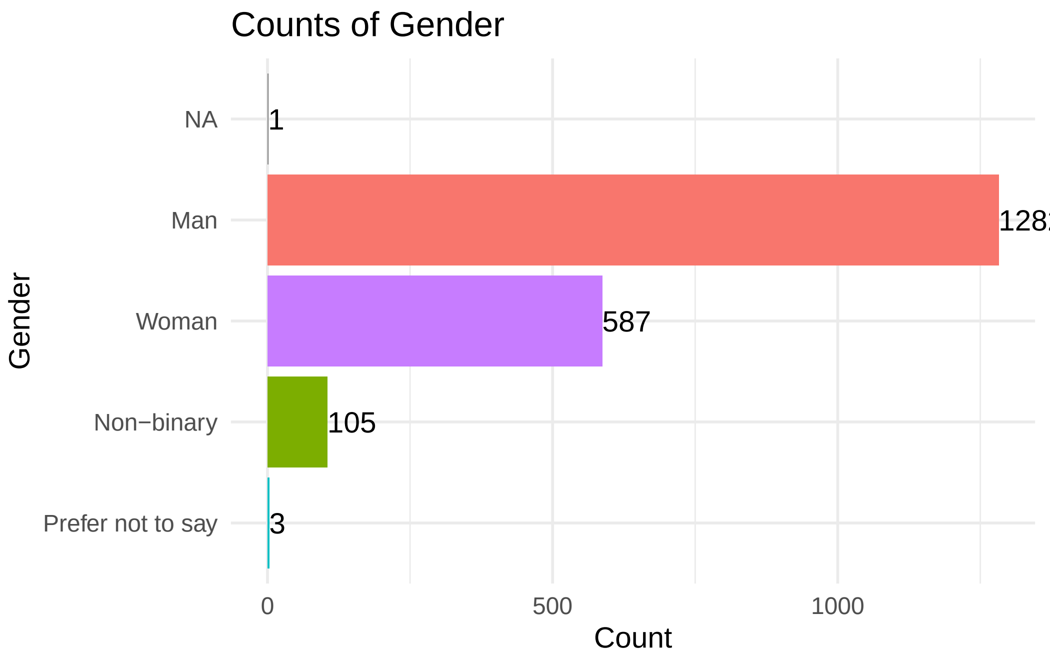
```r
ggplot() +
  geom_rect(
    data = bmi_categories,
    aes(xmin = -Inf, xmax = Inf, ymin = bmi_min, ymax = bmi_max, fill = category),
    alpha = 0.2) +
  geom_boxplot(
    data = intake_participants,
    aes(
      x = fct_infreq(ethnicity),
      y = bmi
    ),
    fill = "white"
    ) +
  scale_fill_manual(values = category_palette, name = "BMI Categories") +
  scale_colour_brewer(palette = "Set2", name = "Ethnicity") +
  labs(
    title = "BMI Distribution Across Gender",
    x = "Ethnicities",
    y = expression(Body ~ Mass ~ Index ~ (kg/m^2))
  ) +
  guides(fill = "legend", colour = "none") +
  coord_flip()
```

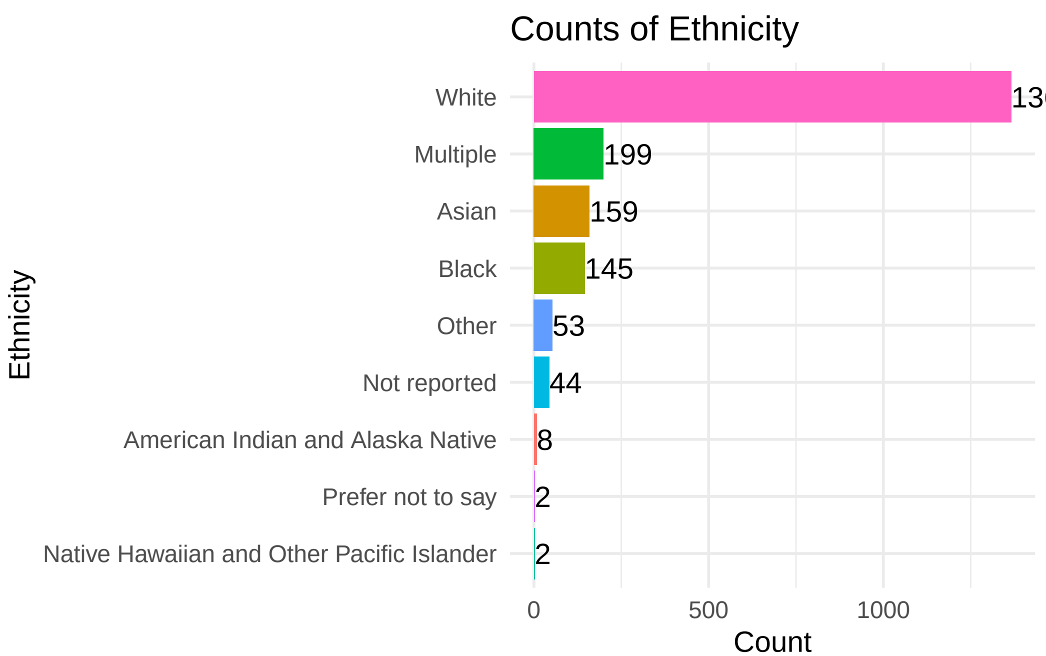


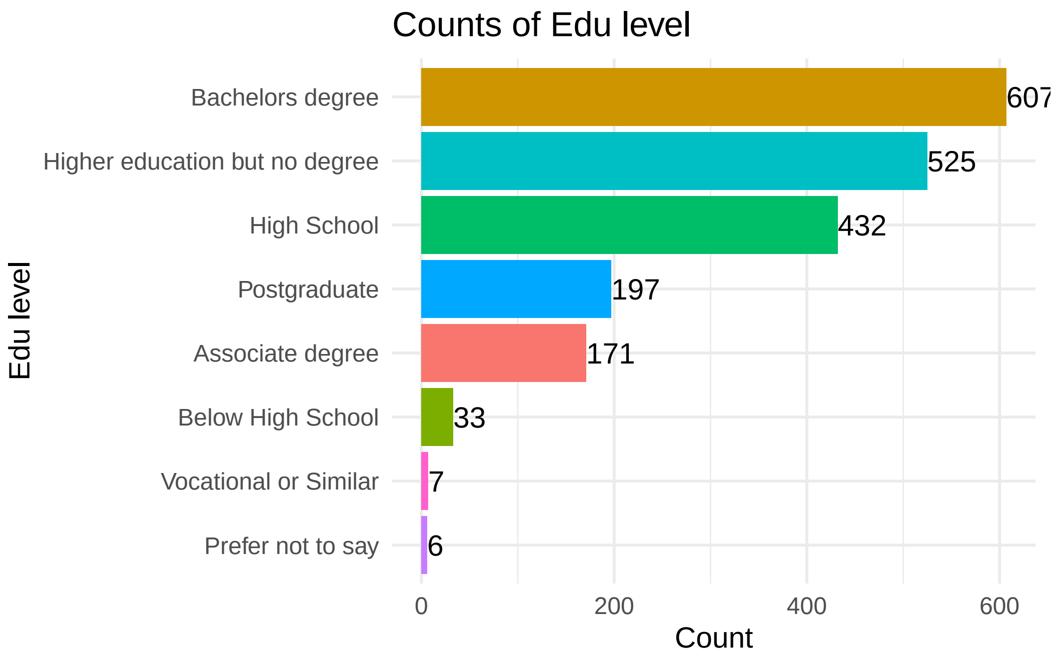
Now for the categoric variables:

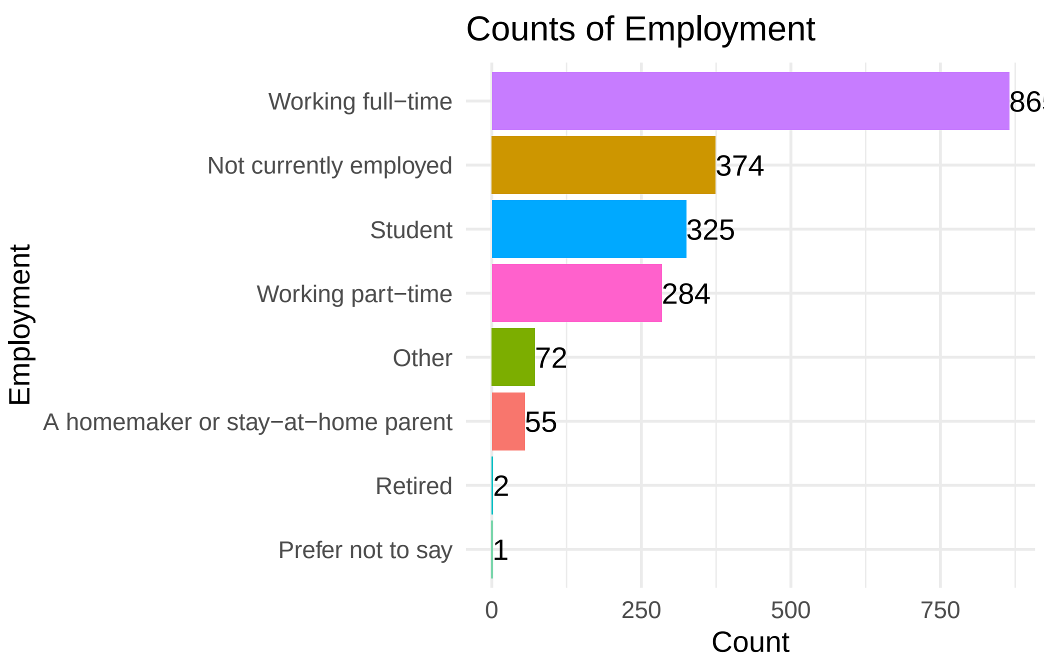
```
for (col in demographic_vars[!demographic_vars %in% c("pid", "geo_area", "height", "weight",
  col_freq <- intake_participants |> count(.data[[col]])
  pretty_col <- str_replace_all(str_to_title(col), "_", " ")
  print(
    ggplot(col_freq, aes(x = n, y = fct_reorder(.data[[col]], n), fill = .data[[col]])) +
    geom_col() +
    coord_cartesian(clip = "off") +
    geom_text(aes(label = n), hjust = 0) +
    labs(
      x = "Count",
      y = pretty_col,
      title = paste("Counts of", pretty_col)
    ) +
    theme(
      legend.position = "none"
    )
  )
}
```

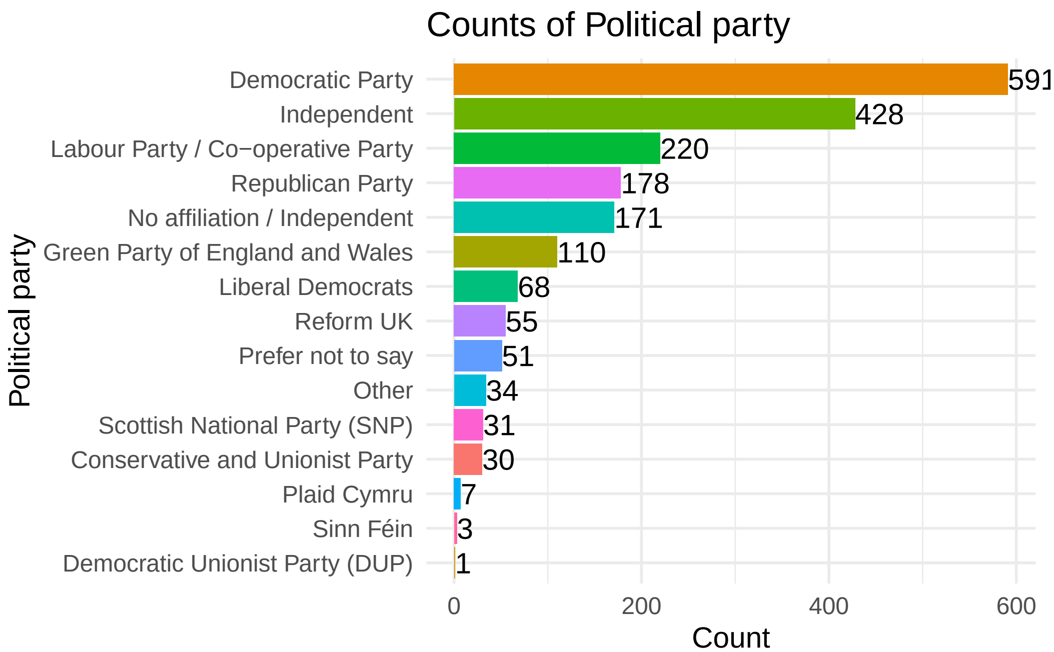


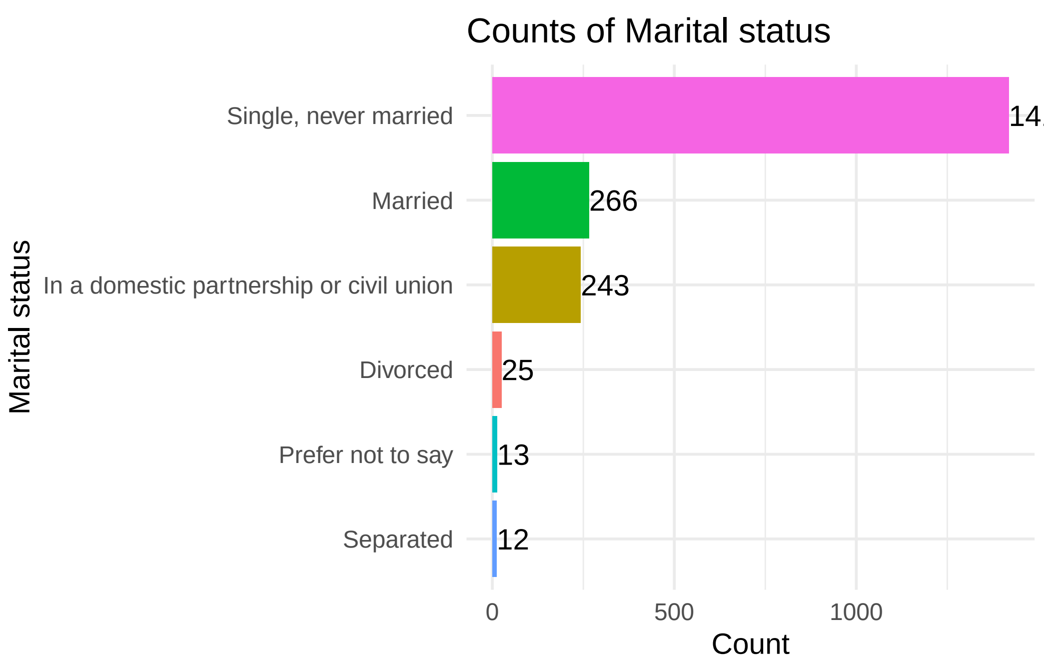

Counts of Gender
Gender
NA
Man
Woman
Non−binary
Prefer not to say
1
128
587
105
3
0
500
1000
Count

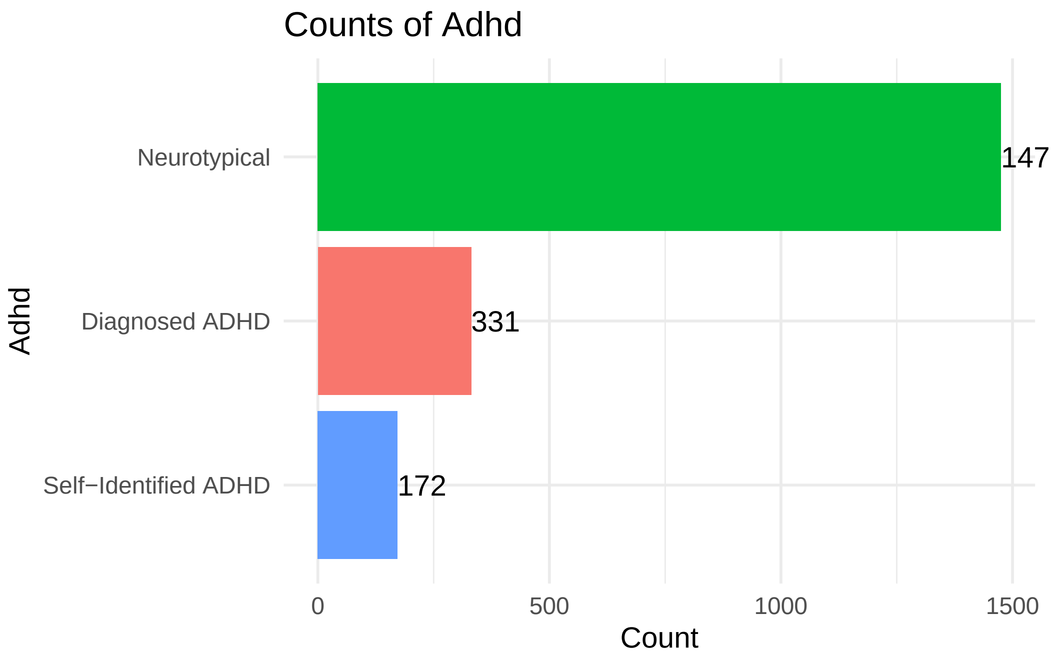

Counts of Ethnicity
Ethnicity
White
Multiple
Asian
Black
Other
Not reported
American Indian and Alaska Native
Prefer not to say
Native Hawaiian and Other Pacific Islander
13
199
159
145
53
44
8
2
2
0
500
1000
Count


Counts of Edu level
Bachelors degree
Higher education but no degree
High School
Postgraduate
Associate degree
Below High School
Vocational or Similar
Prefer not to say
Edu level
607
525
432
197
171
33
7
6
0
200
400
600
Count


Counts of Employment
Working full−time
Not currently employed
Student
Working part−time
Other
A homemaker or stay−at−home parent
Retired
Prefer not to say
Employment
374
325
284
72
55
2
1
0
250
500
750
Count


Counts of Political party
Political party
Democratic Party
Independent
Labour Party / Co−operative Party
Republican Party
No affiliation / Independent
Green Party of England and Wales
Liberal Democrats
Reform UK
Prefer not to say
Other
Scottish National Party (SNP)
Conservative and Unionist Party
Plaid Cymru
Sinn Féin
Democratic Unionist Party (DUP)
591
428
220
178
171
110
68
55
51
34
31
30
7
3
1
0
200
400
600
Count


Counts of Marital status
Marital status
Single, never married
Married
In a domestic partnership or civil union
Divorced
Prefer not to say
Separated
141
266
243
25
13
12
0
500
1000
Count



May be worth considering Independent (the american party) was assumed to be no affiliation by the UK participants:

```
intake_participants |> filter(political_party == "Independent") |> pull(country) |> table()
```

```

 US
428
```

No one from the UK stated this, which is good.

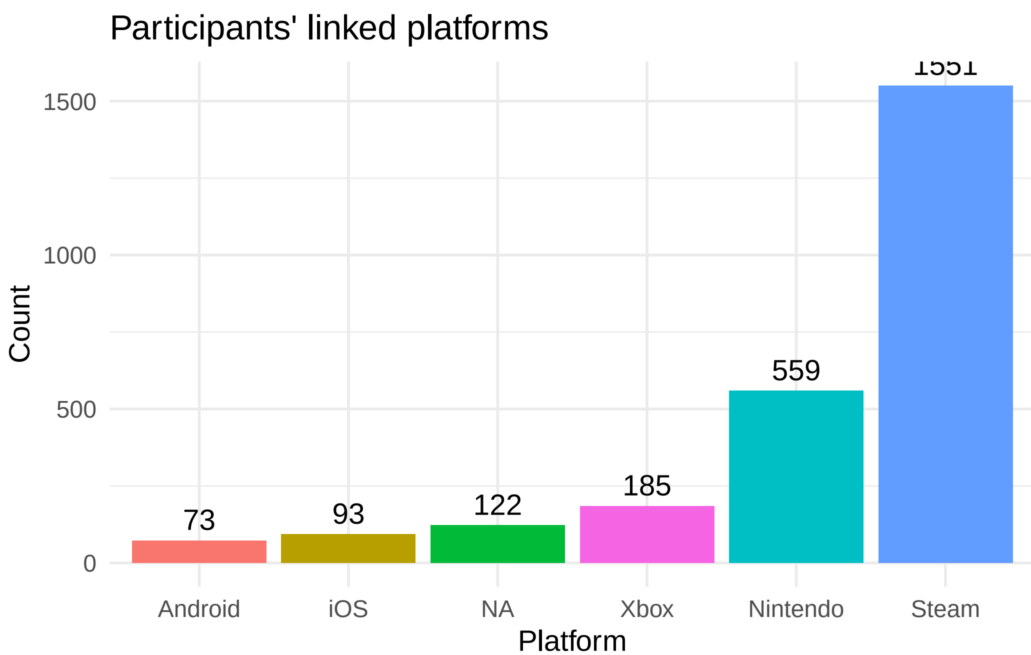
Video Game Statistics

General Descriptive Statistics

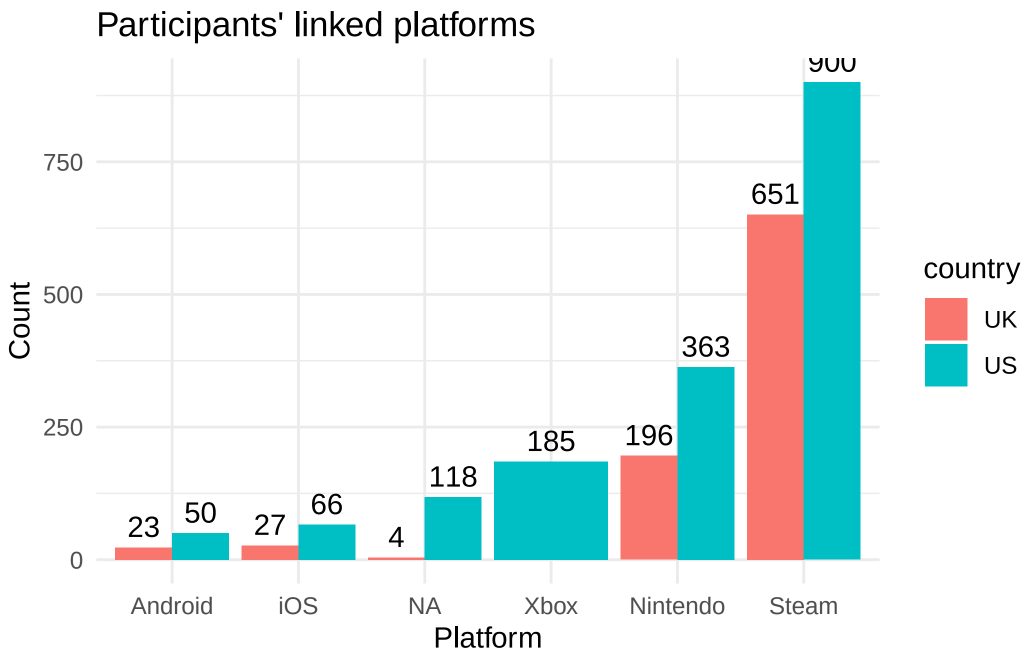
This focuses on more about the general information taken from the participant data (connected platforms, self reported hours, proportion).

```
linked_platform_intake <- intake_participants |>
  select(all_of(c(demographic_vars, "linked_platforms"))) |>
  separate_longer_delim(linked_platforms, delim = ", ") |>
  rename(platform = linked_platforms)
```

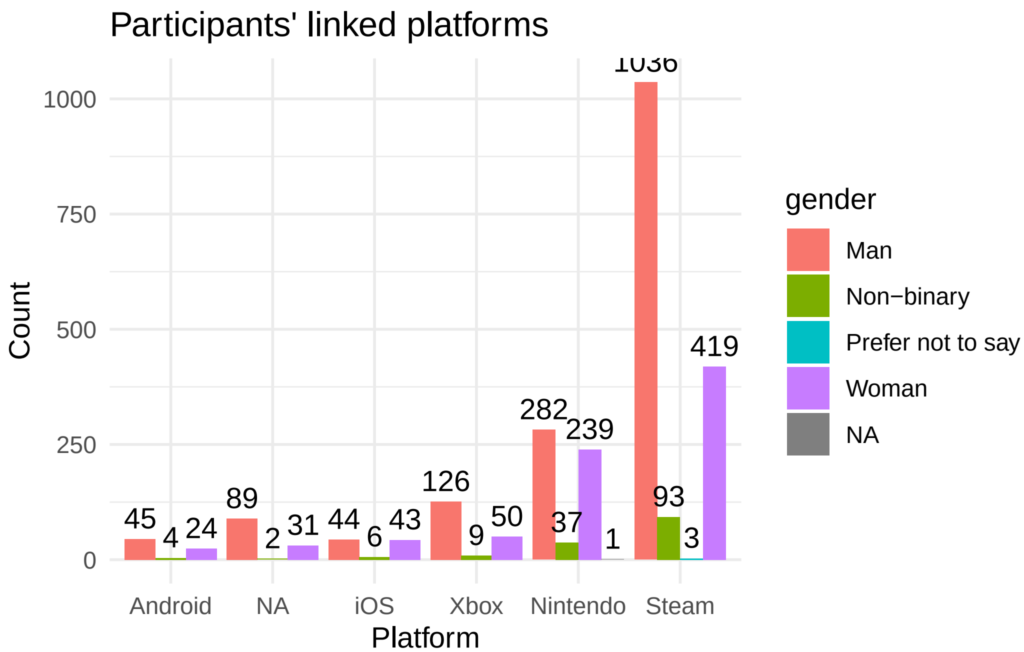
```r
# bar chart of platform frequency
linked_platform_intake |>
  count(platform) |>
  mutate(platform = if_else(is.na(platform), "NA", platform)) |>
  ggplot(aes(x = fct_reorder(platform, n), y = n, fill = platform)) +
  geom_col() +
  geom_text(aes(label = n, vjust = -0.5), position = position_dodge(0.9)) +
  labs(x = "Platform", y = "Count", title = "Participants' linked platforms") +
  theme(legend.position = "none")
```




```r
# by country
linked_platform_intake |>
  count(platform, country) |>
  mutate(platform = if_else(is.na(platform), "NA", platform)) |>
  ggplot(aes(x = fct_reorder(platform, n), y = n, fill = country)) +
  geom_col(position = position_dodge()) +
  geom_text(aes(label = n, vjust = -0.5), position = position_dodge(0.9)) +
  labs(x = "Platform", y = "Count", title = "Participants' linked platforms")
```




```
# by gender
linked_platform_intake |>
  count(platform, gender) |>
  mutate(platform = if_else(is.na(platform), "NA", platform)) |>
  ggplot(aes(x = fct_reorder(platform, n), y = n, fill = gender)) +
  geom_col(position = position_dodge()) +
  geom_text(aes(label = n, vjust = -0.5), position = position_dodge(0.9)) +
  labs(x = "Platform", y = "Count", title = "Participants' linked platforms")
```



Only US participants linked their Xbox account

Now I will look at the platforms average use at different aggregation levels. Some people were qualified participants, but never linked platforms.

```
weekly_platforms <- linked_platform_intake |>
  left_join(weekly_all, by = c("pid", "platform")) |>
  filter(!is.na(week)) # missing week also misses minutes, this is of no use

daily_platforms <- linked_platform_intake |>
  left_join(daily_telemetry, by = c("pid", "platform")) |>
  rename(day = day_local) |>
  filter(!is.na(day)) # missing day also misses minutes, no reason to keep here

message("Summary of weekly platforms:")
```

```
Summary of weekly platforms:
```

```
weekly_platforms |>
  mutate(across(where(is.character), as.factor)) |>
  skim() |> print()
```

```
-- Data Summary ------------------------
```

```
                           Values
Name                       mutate(weekly_platforms, ...
Number of rows             68213
Number of columns          16
_______________________
Column type frequency:
  Date                     1
  factor                   11
  numeric                  4
________________________
Group variables            None

-- Variable type: Date ----------------------------------------------------------
  skim_variable n_missing complete_rate min        max        median
1 week                  0             1 2022-04-24 2025-10-12 2025-03-09
  n_unique
1      182

-- Variable type: factor --------------------------------------------------------
   skim_variable   n_missing complete_rate ordered n_unique
 1 pid                     0         1     FALSE       1816
 2 country                 0         1     FALSE          2
 3 geo_area                0         1     FALSE       1031
 4 gender                 76         0.999 FALSE          4
 5 ethnicity               0         1     FALSE          9
 6 edu_level               0         1     FALSE          7
 7 employment              0         1     FALSE          8
 8 political_party         0         1     FALSE         15
 9 marital_status          0         1     FALSE          6
10 adhd                    0         1     FALSE          3
11 platform                0         1     FALSE          5
   top_counts
 1 p82: 235, p13: 227, p79: 225, p36: 212
 2 US: 47245, UK: 20968
 3 300: 696, 891: 579, 328: 564, 782: 558
 4 Man: 44506, Wom: 19698, Non: 3914, Pre: 19
 5 Whi: 49414, Mul: 7078, Bla: 5092, Asi: 4109
 6 Bac: 21430, Hig: 17500, Hig: 13474, Pos: 7761
 7 Wor: 30209, Not: 14034, Stu: 9301, Wor: 9223
 8 Dem: 23034, Ind: 15629, Rep: 6879, Lab: 6447
 9 Sin: 45545, Mar: 11868, In : 8545, Div: 1402
10 Neu: 50492, Dia: 11932, Sel: 5789
11 Ste: 27296, Xbo: 22015, Nin: 17981, iOS: 529
```

```
-- Variable type: numeric --------------------------------------------------------
  skim_variable n_missing complete_rate  mean     sd    p0   p25   p50    p75
1 height                0             1 174.   10.2  144.  168.  175.   180.
2 weight                0             1  86.0  24.4   31.8  68.0  81.6   99.8
3 age                   0             1  27.4   5.20  18     24    27     30
4 minutes               0             1 847.  922.     0   157   515.  1226.
   p100 hist
1  203.
2  227.
3   40
4 6035
```

```r
message("Summary of daily platforms:")
```

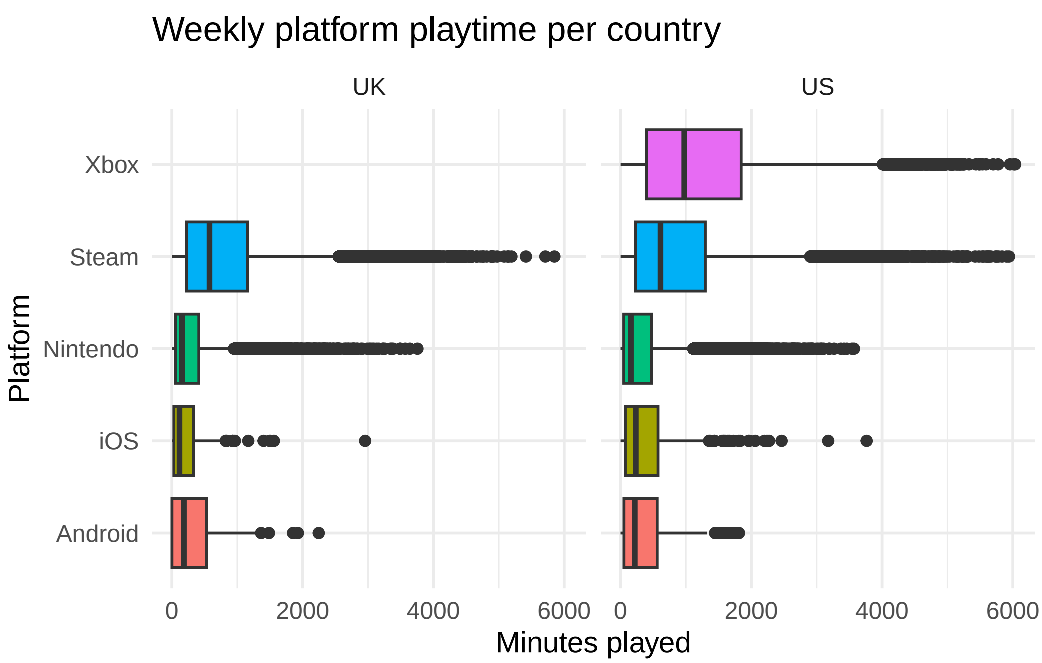
```
Summary of daily platforms:
```

```r
daily_platforms |>
  mutate(across(where(is.character), as.factor)) |>
  skim() |> print()
```

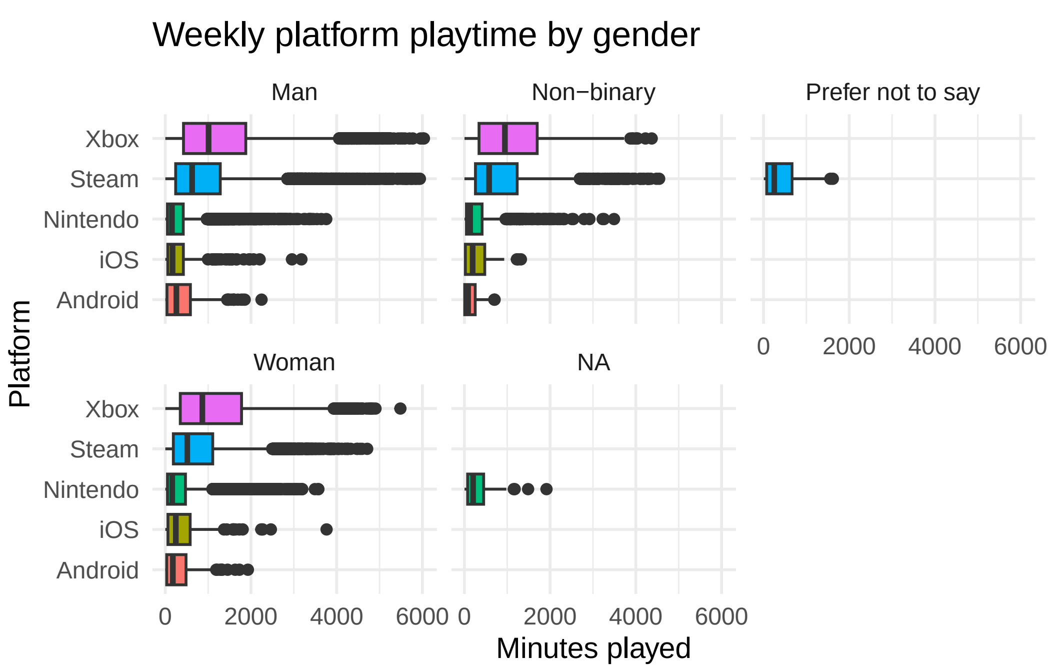
```
-- Data Summary ------------------------
                           Values
Name                       mutate(daily_platforms, a...
Number of rows             289937
Number of columns          16

_______________________
Column type frequency:
  Date                     1
  factor                   11
  numeric                  4

________________________
Group variables            None

-- Variable type: Date -----------------------------------------------------------
  skim_variable n_missing complete_rate min        max        median
1 day                   0             1 2022-04-30 2025-10-17 2025-03-13
  n_unique
1     1267

-- Variable type: factor ---------------------------------------------------------
```

```
   skim_variable   n_missing complete_rate ordered n_unique
 1 pid                     0         1     FALSE       1816
 2 country                 0         1     FALSE          2
 3 geo_area                0         1     FALSE       1031
 4 gender                310         0.999 FALSE          4
 5 ethnicity               0         1     FALSE          9
 6 edu_level               0         1     FALSE          7
 7 employment              0         1     FALSE          8
 8 political_party         0         1     FALSE         15
 9 marital_status          0         1     FALSE          6
10 adhd                    0         1     FALSE          3
11 platform                0         1     FALSE          5
   top_counts
 1 p93: 1351, p36: 1326, p44: 1281, p13: 1269
 2 US: 210905, UK: 79032
 3 328: 3120, 300: 3013, 606: 2493, 339: 2415
 4 Man: 195717, Wom: 78057, Non: 15784, Pre: 69
 5 Whi: 209808, Mul: 31232, Bla: 22057, Asi: 16599
 6 Bac: 87350, Hig: 77085, Hig: 59505, Ass: 30359
 7 Wor: 122730, Not: 67023, Wor: 39487, Stu: 37195
 8 Dem: 104012, Ind: 68908, Rep: 31213, Lab: 23623
 9 Sin: 195225, Mar: 49841, In : 35453, Div: 5918
10 Neu: 214637, Dia: 51625, Sel: 23675
11 Ste: 117030, Xbo: 110650, Nin: 56299, iOS: 3524

-- Variable type: numeric ------------------------------------------------------
  skim_variable n_missing complete_rate  mean     sd    p0   p25   p50   p75
1 height                0             1 175.   10.0  144.  168.  175.  183.
2 weight                0             1  86.9  24.7   31.8  68.0  83.9  99.8
3 age                   0             1  27.5   5.18  18     24    27    30
4 minutes               0             1 199.  185.     0     60   140   285
   p100 hist
1  203.
2  227.
3   40
4 1066.
```

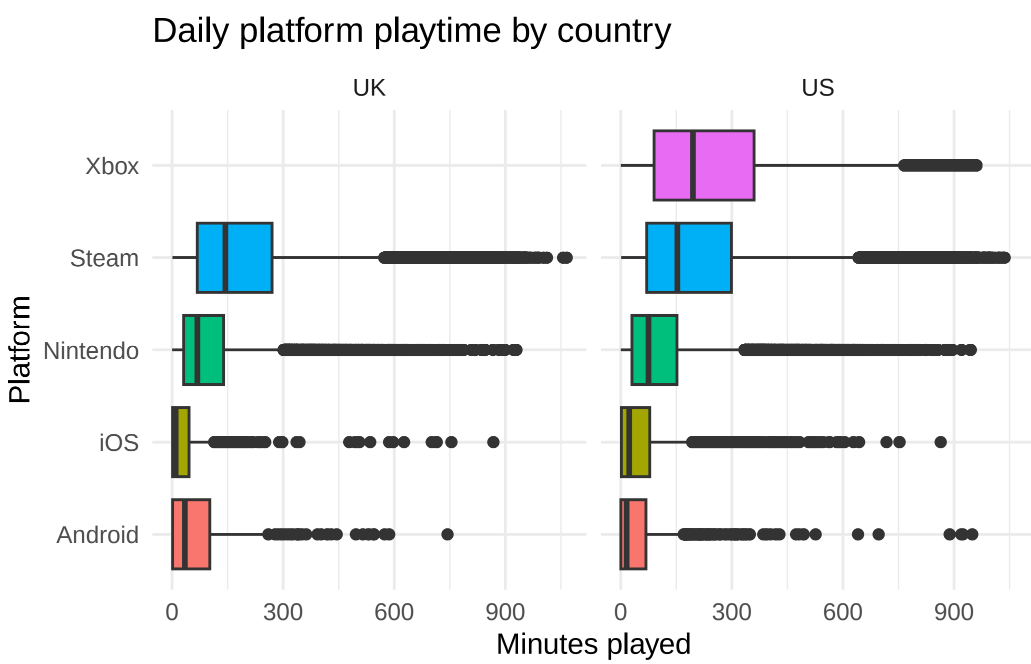
```r
ggplot(weekly_platforms, aes(x = minutes, y = platform, fill = platform)) +
  geom_boxplot() +
  theme(legend.position = "none") +
  facet_wrap(~country) +
  labs(x = "Minutes played", y = "Platform", title = "Weekly platform playtime per country")
```



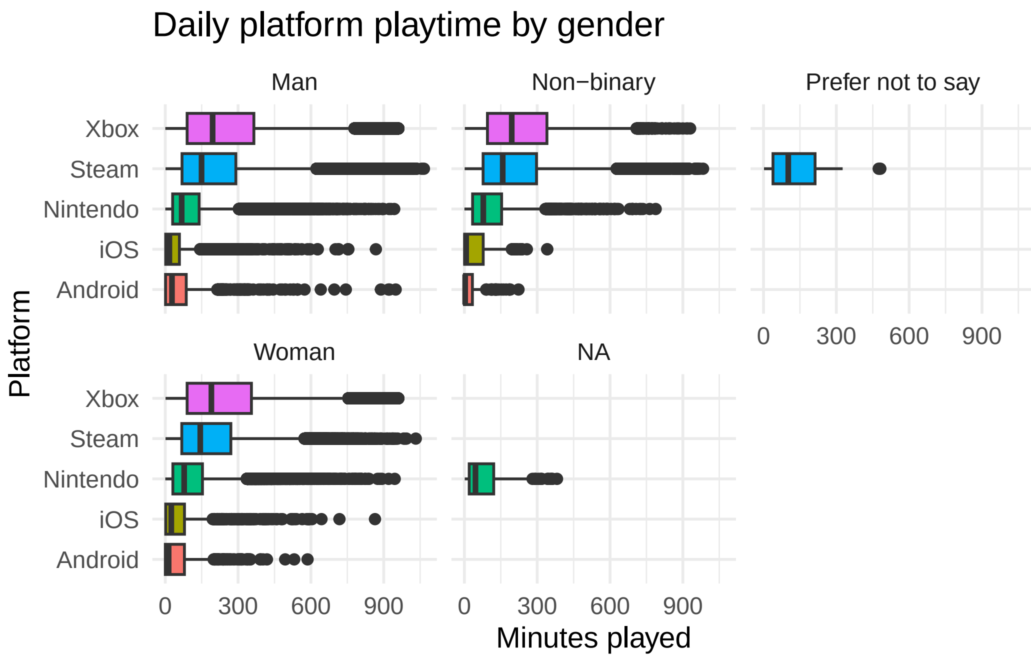

```
ggplot(weekly_platforms, aes(x = minutes, y = platform, fill = platform)) +
  geom_boxplot() +
  theme(legend.position = "none") +
  facet_wrap(~gender) +
  labs(x = "Minutes played", y = "Platform", title = "Weekly platform playtime by gender")
```



```r
ggplot(daily_platforms, aes(x = minutes, y = platform, fill = platform)) +
  geom_boxplot() +
  theme(legend.position = "none") +
  facet_wrap(~country) +
  labs(x = "Minutes played", y = "Platform", title = "Daily platform playtime by country")
```

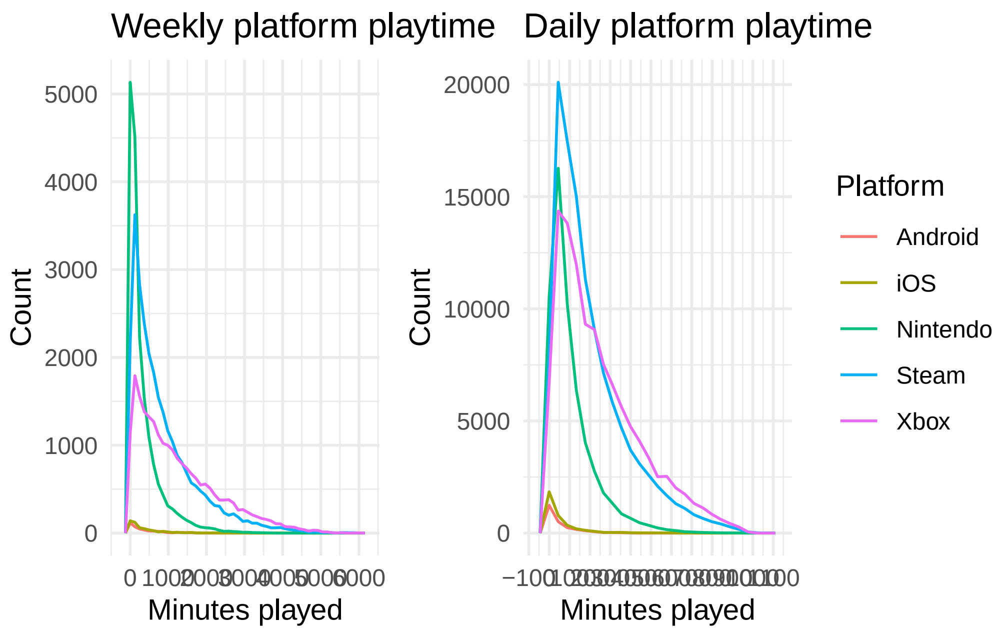



```r
ggplot(daily_platforms, aes(x = minutes, y = platform, fill = platform)) +
  geom_boxplot() +
  theme(legend.position = "none") +
  facet_wrap(~gender) +
  labs(x = "Minutes played", y = "Platform", title = "Daily platform playtime by gender")
```



These boxplots have a lot of outliers, I will try a density plot for a bit better comparisons

```
weekly_counts <- ggplot(weekly_platforms, aes(x = minutes, colour = platform)) +
  geom_freqpoly(bins = 50) +
  scale_x_continuous(breaks = scales::pretty_breaks(n = 10)) +
  labs(
    title = "Weekly platform playtime",
    x = "Minutes played",
    y = "Count",
    colour = "Platform"
  )

daily_counts <- ggplot(daily_platforms, aes(x = minutes, colour = platform)) +
  geom_freqpoly(bins = 25) +
  scale_x_continuous(breaks = scales::pretty_breaks(n = 10)) +
  labs(
    title = "Daily platform playtime",
    x = "Minutes played",
    y = "Count",
    colour = "Platform"
  )

weekly_counts + daily_counts + plot_layout(guides = "collect")
```

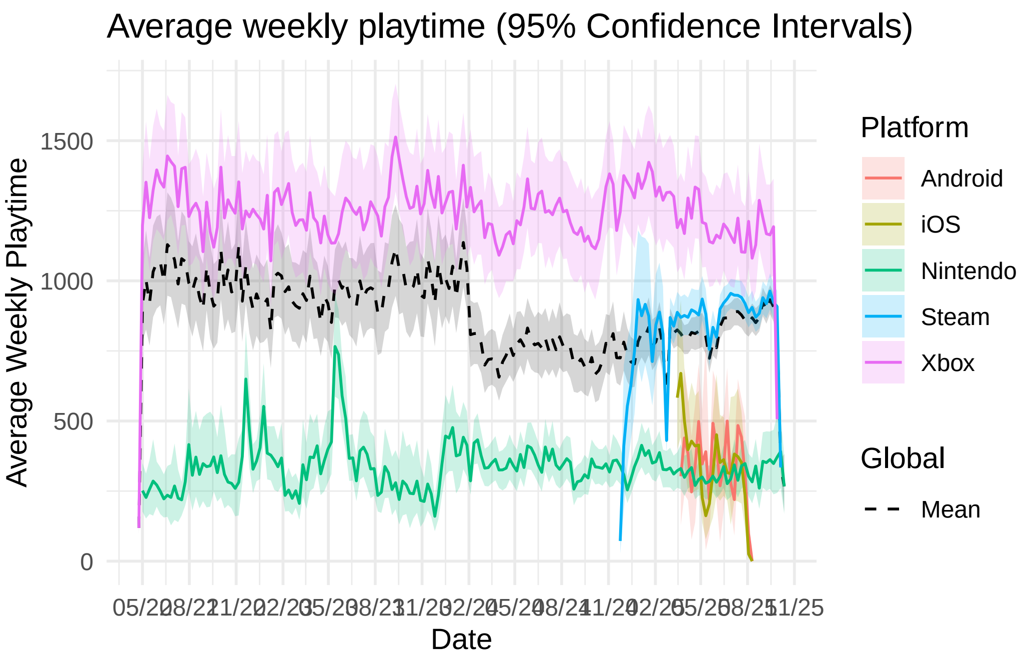


Panels

Now I have a general idea of the platform telemetry, I want to get an idea of the different panels.

```
# 95% confidence intervals
# plot weekly telemetry by platform
ggplot(weekly_platforms, aes(x = week, y = minutes)) +
  # 95% confidence intervals for overall playtime
  stat_summary(aes(group = 1), geom = "ribbon", fun.data = mean_cl_boot,  alpha = 0.2) +
  stat_summary(aes(group = 1, linetype = "Mean"), geom = "line", fun = mean) +
  # 95% confidence intervals for each platform
  stat_summary(aes(group = platform, fill = platform), geom = "ribbon", fun.data = mean_cl_bo
  # mean playtime per platform
  stat_summary(aes(group = platform, colour = platform), geom = "line", fun = mean) +
  scale_linetype_manual(values = c("Mean" = "dashed"), name = "Global") +
  scale_x_date(date_breaks = "3 months", date_labels = "%m/%y") +
  labs(
    title = "Average weekly playtime (95% Confidence Intervals)",
    x = "Date",
    y = "Average Weekly Playtime",
    fill = "Platform",
    colour = "Platform"
  )
```

```
Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom_ribbon()`).
```



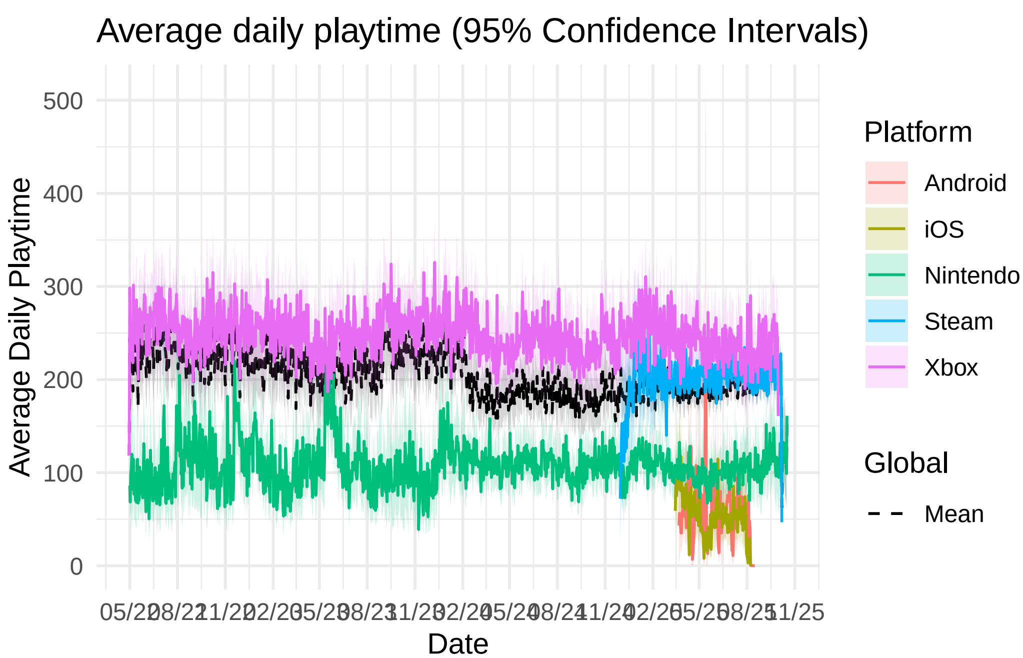
We can Xbox has a higher average weekly playtime, larger than any other platform. Nintendo is consistently lower, which I was slightly surprised with as I expected the portability of the Switch to factor into playtime.

```
# 95% confidence intervals
# plot weekly telemetry by platform
ggplot(daily_platforms, aes(x = day, y = minutes)) +
  # 95% confidence intervals for overall playtime
  stat_summary(aes(group = 1), geom = "ribbon", fun.data = mean_cl_boot,  alpha = 0.2) +
  stat_summary(aes(group = 1, linetype = "Mean"), geom = "line", fun = mean) +
  # 95% confidence intervals for each platform
  stat_summary(aes(group = platform, fill = platform), geom = "ribbon", fun.data = mean_cl_bo
  # mean playtime per platform
  stat_summary(aes(group = platform, colour = platform), geom = "line", fun = mean) +
  scale_linetype_manual(values = c("Mean" = "dashed"), name = "Global") +
  scale_x_date(date_breaks = "3 months", date_labels = "%m/%y") +
  labs(
    title = "Average daily playtime (95% Confidence Intervals)",
    x = "Date",
    y = "Average Daily Playtime",
    fill = "Platform",
```

```
    colour = "Platform"
  )
```

```
Warning: Removed 1 row containing missing values or values outside the scale range
(`geom_ribbon()`).
```

```
Warning: Removed 9 rows containing missing values or values outside the scale range
(`geom_ribbon()`).
```



Biweekly shows this better.

ADHD

I will first create a variable indicating whether a person has ADHD.

```
intake_participants <- intake_participants |>
  mutate(
    adhd = case_when(
      !is.na(neuro_diag_adhd) ~ "Diagnosed ADHD",
      is.na(neuro_diag_adhd) & !is.na(neuro_iden_adhd) ~ "Self-Identified ADHD",
      TRUE ~ "Neurotypical"
```

```
    )
  )

table(intake_participants$adhd)
```

```

      Diagnosed ADHD         Neurotypical Self-Identified ADHD 
                 331                 1475                  172 
```

Now I shall quickly reproduce some of the previous visualisations to see how